

$$C,C^{\#},D,D^{\#},E,F,F^{\#},G,G^{\#},A,A^{\#},B$$

$$\sqrt[12]{2}$$

$$C$$

$$F^{\#}$$

$$\sqrt[6]{2}$$

$$\sqrt{2}$$

$$\sqrt[4]{2}$$

$$\sqrt[2]{2}$$

$$\sqrt[12]{2}$$

$$C$$

$$F^{\#}$$

$$C^{\#} \rightarrow$$

$$D \rightarrow$$

$$D^{\#} \rightarrow$$

$$E \rightarrow$$

$$F \rightarrow$$

$$F^{\#}$$

$$F^{\#} \over f_C = (\sqrt[12]{2})^6$$

$$6/12 =$$

$$2^{1/2} =$$

$$\sqrt{2}$$

$$F^{\#} \over f_C = \sqrt{2}$$

$$\sqrt{2}$$

$$\hline n$$

$$1$$

$$\frac{1}{3}$$

$$\frac{1}{9}$$

$$\frac{1}{9}$$

$$\frac{1}{9}$$

$$\left(\frac{5}{3}\right)^{\frac{n}{2}}$$

$$\left(\frac{5}{3}\right)^n$$

$$\left(\frac{5}{3}\right)^{2n}$$

$$\left(\frac{5}{3}\right)^{n(2n-1)}$$

$$0$$

$$0 =$$

$$0$$

$$1$$

$$5$$

$$\frac{1}{3}$$

$$\frac{1}{3} =$$

$$5 \left(\frac{1}{3}\right) =$$

$$\frac{5}{3}$$

$$\frac{5}{3}$$

$$\frac{5}{3}$$

$$\frac{5}{3}$$

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$$\frac{5}{3}$$

$$2b_2\leq b_1+\frac{1}{2}.$$

$$2^+_3> l_1.$$

$$2^-_1+(1-b_2)> b_1.$$

$$\frac{1}{2}^+_1< b_2< \frac{1}{2}^+_1$$

NOT

$$\omega_0|\omega_0)\approx 0$$

$$\omega_0)|ishighfor\omega<\omega_0and\omega>\omega_0.$$

$$f^{(0)}(x)\neq f'(x)\neq x$$

$$a\neq \frac{b}{a}$$

$$\frac{b}{b}$$

$$\frac{f(a)}{f(b)}>\frac{f(a)}{f(b)}= \frac{f(a)}{f(b)}\neq \frac{f(a)}{f(b)}<$$

$$\frac{f(x)}{\neq 0}for all x.$$

$$\frac{f(x)}{[a,b]}(a,b)$$

$$\frac{c}{(a,b)}\neq 0for all x.$$

$$\frac{f(a)}{f(b)}\neq \frac{f(a)}{f(b)}.$$

$$\frac{(sum.west)+(0.15,0.25)+ (sum.south)+(0.18,0.18)}{(sum.west)+(-1.2,0)}$$

$$G(s)=25s(s+5)$$

$$(sum.south)+(0,-1.3)$$

$$\begin{aligned}
 R_3 &= \\
 2k\Omega, V_3 &= \\
 1V \\
 \pm V_{sat} &= \\
 \pm 15V \\
 \overset{out}{=} & \\
 -\left(\frac{10}{5}(2) + \frac{10}{4}(2) + \frac{10}{2}(1)\right) \\
 \overset{out}{=} & \\
 -(2(2) + 2.5(2) + 5(1)) \\
 \overset{out}{=} & \\
 -(4+ \\
 5+ \\
 5) &= \\
 -14V \\
 -14V \\
 -15V \\
 +15V \\
 \hline
 kHz \\
 R &= \\
 1k\Omega \\
 R_1 &= \\
 2k\Omega \\
 R_2 &= \\
 6k\Omega \\
 C &= \\
 0.1\mu F \\
 - \\
 + \\
 +V_{sat} &= \\
 +15V \\
 -V_{sat} &= \\
 -15V \\
 R_1 \\
 R_2 \\
 V_{out} \\
 C \\
 R \\
 R \\
 C
 \end{aligned}$$

$$\begin{aligned}
 f_0 \\
 R \\
 C &= \\
 0 &= \\
 \frac{1}{2\pi RC} \\
 R &= \\
 1k\Omega \\
 10^3\Omega \\
 C &= \\
 0.1\mu F = \\
 0.1 \times \\
 10^{-6}F = \\
 10^{-7}F \\
 R \\
 C &= \\
 0 &= \\
 \frac{1}{2\pi(10^3)(10^{-7})} = \\
 \frac{1}{2\pi \times 10^{-4}} Hz \\
 0 &= \\
 \frac{10^4}{2\pi} Hz = \\
 \frac{10000}{2\pi} \approx \\
 1591.549Hz \\
 kHz \\
 0 &= \\
 \frac{1591.549}{1000} kHz \approx \\
 1.5915kHz \\
 I_{rms} \\
 \hline
 V_{rms} &= \\
 240V \\
 60\Omega \\
 10mH \\
 8\mu F \\
 I_{rms}
 \end{aligned}$$

$$\begin{aligned}
 v(t) &= \\
 V_m \cos(\omega t) \\
 Ri(t) + L \frac{di(t)}{dt} + \frac{1}{C} \int i(t) dt &= v(t)
 \end{aligned}$$

$$\begin{aligned}
 i(t) &= \\
 I_m \cos(\omega t - \\
 \phi) \\
 V(t) &= \\
 V_{rms} \sqrt{2} e^{j\omega t} \\
 I(t) &= \\
 \sqrt{2} i(t) e^{j\omega t}
 \end{aligned}$$